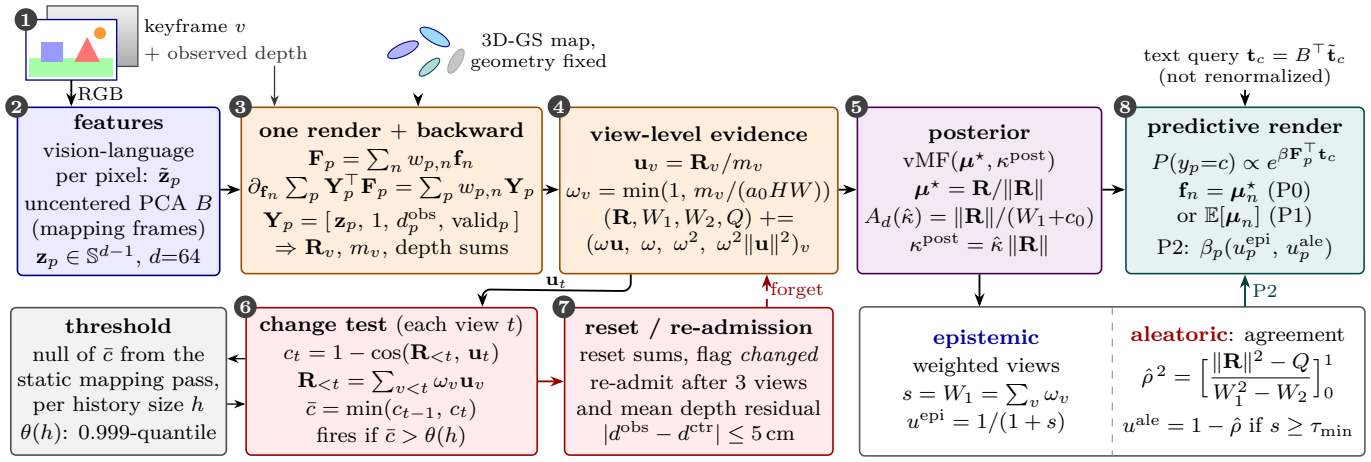
1
keyframe v
+ observed depth
RGB
3D-GS map,
geometry fixed
text query $\mathbf{t}_c = B^\top \tilde{\mathbf{t}}_c$
(not renormalized)
2
features
vision-language
per pixel: $\tilde{\mathbf{z}}_p$
uncentered PCA B
(mapping frames)
$\mathbf{z}_p \in \mathbb{S}^{d-1}$, d=64
3
one render + backward
$\mathbf{F}_p = \sum_n w_{p,n}\mathbf{f}_n$
$\partial_{\mathbf{f}_n} \sum_p \mathbf{Y}_p^\top \mathbf{F}_p = \sum_p w_{p,n}\mathbf{Y}_p$
$\mathbf{Y}_p = [\,\mathbf{z}_p,\ 1,\ d_p^{\text{obs}},\ \text{valid}_p\,]$
$\Rightarrow \mathbf{R}_v,\ m_v$, depth sums
4
view-level evidence
$\mathbf{u}_v = \mathbf{R}_v / m_v$
$\omega_v = \min(1,\ m_v/(a_0 HW))$
$(\mathbf{R}, W_1, W_2, Q)$ +=
$(\omega\mathbf{u},\ \omega,\ \omega^2,\ \omega^2\|\mathbf{u}\|^2)_v$
5
posterior
$\text{vMF}(\boldsymbol{\mu}^\star, \kappa^{\text{post}})$
$\boldsymbol{\mu}^\star = \mathbf{R}/\|\mathbf{R}\|$
$A_d(\hat{\kappa}) = \|\mathbf{R}\|/(W_1 + c_0)$
$\kappa^{\text{post}} = \hat{\kappa}\,\|\mathbf{R}\|$
8
predictive render
$P(y_p{=}c) \propto e^{\beta \mathbf{F}_p^\top \mathbf{t}_c}$
$\mathbf{f}_n = \boldsymbol{\mu}_n^\star$ (P0)
or $\mathbb{E}[\boldsymbol{\mu}_n]$ (P1)
P2: $\beta_p(u_p^{\text{epi}},\ u_p^{\text{ale}})$
$\mathbf{u}_t$
forget
P2
6
change test (each view t)
$c_t = 1 - \cos(\mathbf{R}_{<t},\ \mathbf{u}_t)$
$\mathbf{R}_{<t} = \sum_{v<t} \omega_v \mathbf{u}_v$
$\bar{c} = \min(c_{t-1},\ c_t)$
fires if $\bar{c} > \theta(h)$
7
reset / re-admission
reset sums, flag changed
re-admit after 3 views
and mean depth residual
$|d^{\text{obs}} - d^{\text{ctr}}| \le 5\,\text{cm}$
threshold
null of $\bar{c}$ from the
static mapping pass,
per history size h
$\theta(h)$: 0.999-quantile
epistemic
weighted views
$s = W_1 = \sum_v \omega_v$
$u^{\text{epi}} = 1/(1+s)$
aleatoric: agreement
$\hat{\rho}^2 = \left[\dfrac{\|\mathbf{R}\|^2 - Q}{W_1^2 - W_2}\right]_0^1$
$u^{\text{ale}} = 1 - \hat{\rho}$ if $s \ge \tau_{\min}$